

# **Phase 1 - Front End Test Plan**

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**Github Repository:** <https://github.com/AllanSangle/SQA-Proj-.git>

## **Structure**

This test plan outlines the tests needed to be executed to ensure the requirements of the digital game distribution system's front end are met. Individual test cases represent one session of the system's input, expected output, and expected daily transaction. The tests are organized by subdirectories to group subsets of test cases. Inside the "test\_case" folder are subfolders for the different categories that separate each test case based on what it is being tested. Inside the category folders are subfolders for each test case, which contain success and failure folders.

- *success*: Contains session information that demonstrates the front end behaving incorrectly by either accepting valid input, or rejecting invalid input.
- *failure*: Displays how a session would look if the front end were to fail to meet a requirement and accept invalid input.

Both folders have text file printouts of input, expected output, and a daily transaction of the process. Daily transactions display information about performed functions and can be used to observe if functions behave correctly or not. Located at the root is the "stored\_data" folder which contains available\_games, current\_user\_accounts, and game\_collection text files containing data that's loaded at startup and login to demonstrate the system with active existing users.

## **Current Directory Structure:**

```
|--root/
|--root/stored_data/
    |--root/stored_data/available_games.txt
    |--root/stored_data/current_user_accounts.txt
    |--root/stored_data/game_collection.txt
|--root/test_cases/
    |--root/test_cases/<test_category>/
        |--root/test_cases/<test_category>/<test_name>/
            |--root/test_cases/<test_category>/<test_name>/success/
                |--root/test_cases/<test_category>/<test_name>/success/input.txt
                |--root/test_cases/<test_category>/<test_name>/success/expected_output.txt
                |--root/test_cases/<test_category>/<test_name>/success/daily_transaction.txt
            |--root/test_cases/<test_category>/<test_name>/failure/
                |--root/test_cases/<test_category>/<test_name>/failure/input.txt
                |--root/test_cases/<test_category>/<test_name>/failure/expected_output.txt
                |--root/test_cases/<test_category>/<test_name>/failure/daily_transaction.txt
```

### Functions To Test and Estimated Time:

- Authentication: 10 min
- Account Creation: 20 min
- Account Deletion: 20 min
- Buying Games: 20 min
- Selling Games: 20 min
- Refunding: 20 min
- Adding Credit: 15 min

### Assumptions:

- Input is printed in one line
- Daily\_transaction remains empty if functions fail
- The front end doesn't update the input files (available\_games, current\_user\_accounts, and game\_collection).
  - New users, new games, deleted users, deleted games, and change in credit will appear updated and remain in new sessions when the back end can run
- Transactions that would result in negative balance, or exceeding the credit limit causes an error
  - Credits exceeding limit of transaction formatting gets cut off

### How to run tests:

Each test will use its own starting inputs so as to not interfere with other tests. Once the test has been runned it will create another folder within the test case file named Actual\_Output. Every subsequent output will create a new Actual\_Output file to document which implementations fit the best for the test. These files will be used to compare the output file to the actual\_output. This process will proceed for every test case file. An example of a success output is shown below.

Once the front end is executable a powershell script will run every test case in numbered order. The script will

1. Create a working directory and copy the input.txt information data into it
2. Launch the front end using the inputted information
3. Supply the test commands through cmd input
4. Capture the console output error output and the daily transaction file created
5. Compare between the expected and actual output and save the difference in a "results" folder within the corresponding test case's directory
6. Continue to the next test case

This process will proceed for every test case file. An example of a successful test run output directory is shown below.

|--root/

-----root/test\_cases

-----root/test\_cases/Authentication/

-----root/test\_cases/Authentication/T01/

-----root/test\_cases/Authentication/T01/success/

-----root/test\_cases/Authentication/T01/success/input.txt

-----root/test\_cases/Authentication/T01/success/expected\_output.txt

-----root/test\_cases/Authentication/T01/success/daily\_transaction.txt

-----root/test\_cases/Authentication/T01/failure/

-----root/test\_cases/Authentication/T01/failure/input.txt

-----root/test\_cases/Authentication/T01/failure/expected\_output.txt

-----root/test\_cases/Authentication/T01/failure/daily\_transaction.txt

-----root/test\_cases/Authentication/T01/Results/

-----root/test\_cases/Authentication/T01/Results/failure/

-----root/test\_cases/Authentication/T01/Results/failure/Actual\_output.txt

-----root/test\_cases/Authentication/T01/Results/failure/Actual\_daily\_transaction.txt

-----root/test\_cases/Authentication/T01/Results/success/

-----root/test\_cases/Authentication/T01/Results/success/Actual\_output.txt

-----root/test\_cases/Authentication/T01/Results/success/Actual\_daily\_transaction.txt

-----root/test\_cases/Buying Games/

-----root/test\_cases/Buying Games/T22

-----root/test\_cases/Buying Games/T22/success

-----root/test\_cases/Buying Games/T22/success/input.txt

-----root/test\_cases/Buying Games/T22/success/expected\_output.txt

-----root/test\_cases/Buying Games/T22/success/daily\_transaction.txt

-----root/test\_cases/Buying Games/T22/failure

-----root/test\_cases/Buying Games/T22/failure/input.txt

-----root/test\_cases/Buying Games/T22/failure/expected\_output.txt

-----root/test\_cases/Buying Games/T22/failure/daily\_transaction.txt

-----root/test\_cases/Buying Games/T22/Results

-----root/test\_cases/Buying Games/T22/Results/failure

-----root/test\_cases/Buying Games/T22/Results/failure/Actual\_output.txt

-----root/test\_cases/Buying Games/T22/Results/failure/Actual\_daily\_transaction.txt

-----root/test\_cases/Buying Games/T22/Results/success

-----root/test\_cases/Buying Games/T22/Results/success/Actual\_output.txt

-----root/test\_cases/Buying Games/T22/Results/success/Actual\_daily\_transaction.txt